

Bridging Theory and Practice: Ethical AI User Story Generator - Translating Ethical AI Requirements into Ethical User Stories

João Rossi
joaorossiborba@gmail.com
University of Brasília (UnB),
Department of Computer Science
Brasília, Brazil

Fabiana Mendes
fabiana.freitas.mendes@gmail.com
University of Brasília (UnB),
Department of Computer Science
Brasília, Brazil

José Siqueira de Cerqueira
jose.siqueiradecerqueira@tuni.fi
Tampere University, Faculty of
Information Technology and
Communication Sciences
Tampere, Finland

Pekka Abrahamsson
pekka.abrahamsson@tuni.fi
Tampere University
Faculty of Information Technology
and Communication Sciences
Tampere, Finland

Edna Dias Canedo
ednacanedo@unb.br
University of Brasília (UnB),
Department of Computer Science
Brasília, Brazil

Abstract

While the subject of ethical guidelines for AI is widely discussed, there is often a lack of practical application in the context of software development. In order to address this issue, we present *EAI-USG*, a tool that automatically translates high-level ethical requirements into Ethical User Stories, transforming ethical principles into concrete tasks within agile workflows. Powered by large language models with retrieval-augmented generation, the tool produces structured stories that practitioners perceive as clear and useful. A preliminary evaluation involving 30 software professionals suggests that this approach facilitates the operationalization of ethics during requirements engineering with only a minor additional effort.

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1 Motivation and Problem

AI has expanded rapidly across research and industry [5, 9], intensifying concerns about its ethical implications [12] and the challenge of balancing societal benefits with the risk of harm [11]. These concerns are underscored by high-profile ethical failures [14], including discriminatory hiring tools [14] and privacy and impersonation abuses [10].

Despite this growing awareness, a persistent challenge lies in the operationalization of ethical principles during the software development process [2, 15]. Although many ethical guidelines have been proposed [4], they often remain abstract and lack practical

alignment with real-world software processes [3, 9], limiting their integration into the software development lifecycle [3, 12]. This is further intensified by the lack of clear regulatory and methodological guidance, as well as the inherent complexity of the domain itself [1]. Moreover, existing tools frequently focus on narrow subsets of ethical principles, further highlighting the need for practical methods that development teams can realistically use [13].

One promising way to address this gap is to translate ethical requirements into actionable artifacts that naturally fit established development workflows. Ethical User Stories (*EUS*) provide such a bridge by expressing abstract ethical principles as concrete, user-centered work items that support early requirements activities [8]. Building on this insight, we introduce the Ethical AI User Story Generator (*EAI-USG*), a tool that automatically translates high-level ethical AI requirements into *EUS*, enabling earlier and more practical integration of ethics into requirements engineering.

2 The *EAI-USG* Tool

We propose *EAI-USG*, a tool that automatically translates high-level ethical AI requirements into *EUS*. Given one or more ethical requirements as input, *EAI-USG* generates structured user stories consisting of a title, a user-centric description following the pattern “As a <persona> I want to <do something> <so that>”, and a set of acceptance criteria expressed as actionable work items.

EAI-USG is formalized as an LLM-based system that converts high-level AI ethical requirements into ethical user stories. It aims to assist software development teams in the practical incorporation of ethical principles into their AI systems. The tool employs RAG techniques [7] and fine-tuning [6] to generate predictions. By incorporating RAGs into the system, it combines the use of a language model with the retrieval of information from specific databases and documents [7], thereby improving the precision and contextualization of the generated models.

The resulting pipeline is lightweight, reproducible, and compatible with agile requirements engineering practices, enabling ethical considerations to be addressed alongside functional and other non-functional requirements during early development stages. An



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example of a generated output is shown in *EUS* 1, derived from the ethical requirement: “The AI system must refrain from making decisions without obtaining explicit user consent.”

Ethical User Story 1

Title: User Consent for Decision Making

Description: As a user, I want the AI to obtain my consent before making decisions on my behalf so that I maintain control over my decisions and ensure my autonomy.

Work:

- (1) Implement a mechanism for users to provide consent for AI to make decisions on their behalf.
- (2) Ensure that the AI does not initiate any action without obtaining prior consent from the user.
- (3) Provide clear communication about what decisions the AI will make and when it requires consent.
- (4) Develop a user interface that allows users to easily revoke or modify their consent settings at any time.

3 Preliminary Results

We conducted an initial validation with 30 practitioners from industry, including developers, requirements engineers, testers, and analysts. Participants interacted with the tool in a guided setting and evaluated the generated *EUS* using a mixed-methods questionnaire composed of Likert-scale and open-ended questions covering usefulness, clarity, coherence, and ethical alignment.

The results indicate a positive overall perception of the generated stories. Most participants considered the outputs to be clear, coherent, and useful for understanding and reasoning about ethical AI requirements. In particular, respondents reported that *EAI-USG* reduced the cognitive and manual effort required to translate abstract ethical principles into concrete development tasks. Several participants highlighted that the use of a familiar user story format helped communication and discussion of ethical concerns within development teams. Qualitative feedback mostly characterized the tool as efficient and innovative, emphasizing its potential to support ethical considerations without disrupting existing workflows.

At the same time, the evaluation revealed relevant limitations. Participants noted that coverage of ethical principles was uneven, with some principles being less consistently represented in the generated stories. In addition, the validation relied primarily on perception-based measures collected in a short-term setting. The study did not include objective metrics, nor did it observe the use of *EAI-USG* within real development teams. These results suggest that while the approach is promising, further empirical evidence is required to assess its effectiveness in practice.

4 Discussion and Next Steps

The preliminary findings suggest that *EAI-USG* can support the operationalization of ethical AI requirements by automatically generating *EUS* that practitioners perceive as clear and actionable. By expressing ethical principles in a familiar requirements artifact, the tool helps to link abstract ethical guidance and everyday software engineering activities, particularly during early stages of requirements elicitation and analysis.

As future work, we plan to focus on improving the underlying language models by adopting more recent state-of-the-art architectures and refining the generation pipeline to enhance output quality and ethical alignment. Furthermore, we aim to strengthen the evaluation by involving a wider and more diverse set of users, enabling results that are more representative and generalizable. This will be complemented by more systematic analysis techniques, such as explicit coverage mapping between generated user stories and ethical principles, to provide a more robust assessment of *EAI-USG*.

5 Artifact Availability

The implementation and supporting artifacts are publicly available on Github and Zenodo.

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